

UFUG 2106 Discrete Mathematics

Assignment 3 – Complete Solutions

Question 1 (10 marks)

Proposition: If a positive integer n is composite, then n has a prime divisor not greater than $\sqrt{n}$.

Proof:

1. Since n is composite, there exist positive integers a, b such that

$$n = a \cdot b, \quad 1 < a \leq b < n.$$

2. Suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Then

$$a \cdot b > \sqrt{n} \cdot \sqrt{n} = n,$$

contradicting $n = a \cdot b$.

3. Hence at least one factor (say a) satisfies $a \leq \sqrt{n}$.

4. Any prime factor of a is $\leq a \leq \sqrt{n}$, so n has a prime divisor $\leq \sqrt{n}$.

Q.E.D.

Question 2 (30 marks)

2.1 Trial Division Primality Test (Pseudocode)

Input: positive integer n

Output: True (prime) / False (not prime)

1. If $n \leq 1$: return False .
2. If $n = 2$: return True .
3. If n is even: return False .
4. For i from 3 to $\lfloor \sqrt{n} \rfloor$, step 2:

- If $n \bmod i = 0$: return False .

5. Return True .

2.2 Mersenne Primes Below $2^{40} - 1$

Definition: If p is prime and $2^p - 1$ is also prime, then $2^p - 1$ is a Mersenne prime.

From

$$2^p - 1 \leq 2^{40} - 1,$$

we get $p \leq 40$.

Python implementation is extracted to: UFUG2106_DiscreteMathematics_Assignment3_Code.py

Result: valid prime exponents p are

2, 3, 5, 7, 13, 17, 19, 31.

Question 3 (30 marks)

3.1 Sieve of Eratosthenes (Pseudocode)

Input: positive integer n

Output: all primes $\leq n$

1. Create boolean array `is_prime[0..n]` , initialized to True ; set `is_prime[0]=is_prime[1]=False` .
2. For i from 2 to $\lfloor \sqrt{n} \rfloor$:
 - If `is_prime[i]` is True , mark $i^2, i^2 + i, \dots \leq n$ as False .
3. Collect all i with `is_prime[i]=True` and return.

3.2 Prime-Counting Function $\pi(n)$

Python implementation is extracted to: UFUG2106_DiscreteMathematics_Assignment3_Code.py

Result Table

n	$\pi(n)$	$n / \ln n$	Ratio	Time (ms)
100	25	21.71	1.1516	~0.01
1000	168	144.76	1.1605	~0.03
10000	1229	1085.74	1.1319	~0.10
100000	9592	8685.89	1.1043	~0.50
1e6	78498	72382.41	1.0845	~3.00
1e7	664579	620420.69	1.0712	~25.00
1e8	5761455	5428681.03	1.0613	~220.00

Conclusion:

$$\pi(n) \sim \frac{n}{\ln n}$$

(Prime Number Theorem), and the ratio approaches 1.

Question 4 (10 marks)

Proposition: For any positive integers a, b ,

$$\gcd(a, b) \cdot \text{lcm}(a, b) = a \cdot b.$$

Proof:

1. Let $d = \gcd(a, b)$, then

$$a = da', \quad b = db', \quad \gcd(a', b') = 1.$$

2. The least common multiple is

$$\text{lcm}(a, b) = da'b'.$$

3. LHS:

$$\gcd(a, b) \cdot \text{lcm}(a, b) = d \cdot (da'b') = d^2 a' b'.$$

4. RHS:

$$a \cdot b = (da')(db') = d^2 a' b'.$$

Hence LHS = RHS, so the identity holds. Q.E.D.

Question 5 (10 marks)

Using the Extended Euclidean Algorithm, find $x, y \in \mathbb{Z}$ such that

$$\gcd(2024, 1960) = 2024x + 1960y.$$

Euclidean algorithm steps:

1. $2024 = 1 \times 1960 + 64$
2. $1960 = 30 \times 64 + 40$
3. $64 = 1 \times 40 + 24$
4. $40 = 1 \times 24 + 16$
5. $24 = 1 \times 16 + 8$
6. $16 = 2 \times 8 + 0$

So $\gcd(2024, 1960) = 8$.

Backward substitution:

$$\begin{aligned} 8 &= 24 - 1 \times 16 \\ &= 24 - 1 \times (40 - 1 \times 24) \\ &= 2 \times 24 - 1 \times 40 \\ &= 2 \times (64 - 1 \times 40) - 1 \times 40 \\ &= 2 \times 64 - 3 \times 40 \\ &= 2 \times 64 - 3 \times (1960 - 30 \times 64) \\ &= 92 \times 64 - 3 \times 1960 \\ &= 92 \times (2024 - 1 \times 1960) - 3 \times 1960 \\ &= 92 \times 2024 - 95 \times 1960. \end{aligned}$$

Therefore,

$$8 = 92 \times 2024 - 95 \times 1960,$$

so $x = 92, y = -95$.

Question 6 (10 marks)

Use the construction in the proof of the Chinese remainder theorem to find all solutions to the system of congruences:

$$\begin{cases} x \equiv 1 \pmod{3} \\ x \equiv 1 \pmod{5} \\ x \equiv 3 \pmod{7} \\ x \equiv 2 \pmod{11} \end{cases}$$

Show all your steps.

Solution:

1. Moduli are pairwise coprime:

$$m_1 = 3, m_2 = 5, m_3 = 7, m_4 = 11.$$

2. Total modulus:

$$M = 3 \times 5 \times 7 \times 11 = 1155.$$

3. Components $M_k = M/m_k$:

$$M_1 = 385, \quad M_2 = 231, \quad M_3 = 165, \quad M_4 = 105.$$

4. Find inverses y_k of $M_k \pmod{m_k}$:

- $385y_1 \equiv 1 \pmod{3} \implies 1y_1 \equiv 1 \pmod{3} \implies y_1 = 1.$
- $231y_2 \equiv 1 \pmod{5} \implies 1y_2 \equiv 1 \pmod{5} \implies y_2 = 1.$
- $165y_3 \equiv 1 \pmod{7} \implies 4y_3 \equiv 1 \pmod{7} \implies y_3 = 2$ (since $4 \times 2 = 8 \equiv 1 \pmod{7}$).
- $105y_4 \equiv 1 \pmod{11} \implies 6y_4 \equiv 1 \pmod{11} \implies y_4 = 2$ (since $6 \times 2 = 12 \equiv 1 \pmod{11}$).

5. Construct solution:

$$\begin{aligned}
x &\equiv a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3 + a_4M_4y_4 \pmod{1155} \\
&\equiv 1(385)(1) + 1(231)(1) + 3(165)(2) + 2(105)(2) \pmod{1155} \\
&\equiv 385 + 231 + 990 + 420 \pmod{1155} \\
&\equiv 2026 \pmod{1155} \\
&\equiv 871 \pmod{1155}.
\end{aligned}$$

General solution:

$$x = 871 + 1155k, \quad k \in \mathbb{Z}.$$